Joshua Ibrom

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TECHNICAL SKILLS

Systems & Languages Rust · Python · Go · Zig · C · TypeScript · PHP · Bash
Infrastructure & Systems Linux Systems & POSIX · Multithreading & Concurrency · TCP/IP & Sockets · Docker & Compose · CI/CD Pipelines · Git
Databases & Storage PostgreSQL · MariaDB · SQLite · Vector Search / RAG · Relational Modeling
Backend & Developer Tooling REST APIs · Laravel · Flask · CLI / TUI Development · Automated Testing

WORK EXPERIENCE

Texas A&M University-San Antonio — San Antonio, TX

Web Applications Developer — *ITS Web Services* — Jul 2025 – Present

- **Engineered a 23x performance gain** (380s to 16s) by rewriting a legacy single-threaded data intake pipeline in Python, eliminating TLS handshake churn via **thread-level TCP connection pooling** and **concurrent I/O**.
- **Architected a multithreaded extraction engine in Python**, achieving a **10x throughput increase** using `ThreadPoolExecutor` and decorator-driven exponential backoff with retry handling to generate structured RAG datasets.
- **Spearheaded automated testing and CI/CD pipelines**, standardizing deployment workflows across the team and replacing manual release bottlenecks.
- **Mentored 4 student developers** via structured code reviews and system architecture demos, overseeing critical intake refactors for high-reliability data processing.

Research Technician — *Office of Research* — Dec 2024 – Jul 2025

- **Engineered an automated data extraction pipeline** scraping HB2504 public endpoints to harvest faculty records across **~650 CVs**, integrating an LLM parsing layer with **SQLite** for transactional persistence.
- **Integrated academic publication tracking** in collaboration with university stakeholders, automating schema updates to streamline institutional visibility of research outputs.

Web Development Intern — *ITS Web Services* — May 2024 – Aug 2024

- **Remediated critical Broken Access Control (BFLA/IDOR) vulnerabilities** in a legacy LAMP application, preventing unauthorized CRUD execution on sensitive records for **8,000+ accounts**.
- **Engineered custom input validation and sanitization layers** to enforce strict type constraints on untrusted request parameters, hardening legacy database queries against injection vectors.

FEATURED PROJECTS

struktur *(Developer Tooling & Systems)* `Rust` `SQLite` `CLI` `Tera` — Ongoing

- Architected a local-first application tracking and document generation suite in **Rust** using a **multi-crate Cargo workspace** (`core`, `cli`, `tui`).
- Engineered embedded persistence via **SQLite (rusqlite)** featuring compile-time embedded migrations (`PRAGMA user_version`), atomic transactions, and cascading audit trails.
- Designed domain schemas enforcing **referential integrity at load-time** via Serde `TryFrom` validation pipelines, preventing invalid state representation in memory.
- Integrated a dynamic templating engine (**Tera**) with context resolution, hook pre-rendering, and cross-platform output routing (stdout, disk, clipboard, WSL).

Academic Knowledge Graph & Discovery Platform *(Backend & Search Infrastructure)* `Laravel` `MariaDB` `Python` — Ongoing

- Consolidating thousands of disparate academic publication records, DOIs, and grants into a queryable relational schema linking faculty across institutional departments.
- Engineered a taxonomy normalization pipeline mapping unconstrained LLM entity extractions to the **Digital Commons Disciplines Taxonomy (Tier 2)**, converting noisy text into structured research clusters.
- Evaluated full-text search indexing strategies in **MariaDB**, balancing query latency against operational complexity to maintain a low-maintenance production footprint.

Squelch *(Systems Programming)* `Zig` `POSIX` `mmap` — Ongoing

- Architected an experimental static site generator in **Zig** achieving absolute zero-allocation parsing by relying solely on pointer slices into **memory-mapped files (`std.posix.mmap`)**.
- Investigated compile-time stream composition techniques to eliminate dynamic heap allocations and intermediate I/O buffers during document generation.

EDUCATION

MS Computer Science — Texas A&M University-San Antonio `GPA: 3.9 / 4.0` — Aug 2024 – May 2026
BS Computer Science — Texas A&M University-San Antonio — Aug 2017 – May 2024

LEADERSHIP & SERVICE

Staff Advisor — TAMU-SA ACM — Jan 2026 – Present

- Developed and led an **Introduction to Rust** workshop, instructing students on **memory safety, borrow checker semantics, and low-level systems concepts**.
- Conducted a **Developer Workflows with Git** session, training students on interactive rebasing, trunk-based branching strategies, and conflict resolution.